

Phase 1: Clustering-Oriented EDA

Records: 283,726

Clustering features: 31

Selected EDA features:

V1, V2, V3, V4, V10, V14, V17, Amount_log1p

External class distribution (not used for clustering):

Class

0	283253
1	473

Clustering Tendency Assessment

Hopkins repeated results

	mean	std	min	max
scaler				
robust	0.999999	0.000001	0.999997	1.0
standard	0.999957	0.000096	0.999785	1.0

Interpretation guide:

H near 0.5 indicates random-like structure.

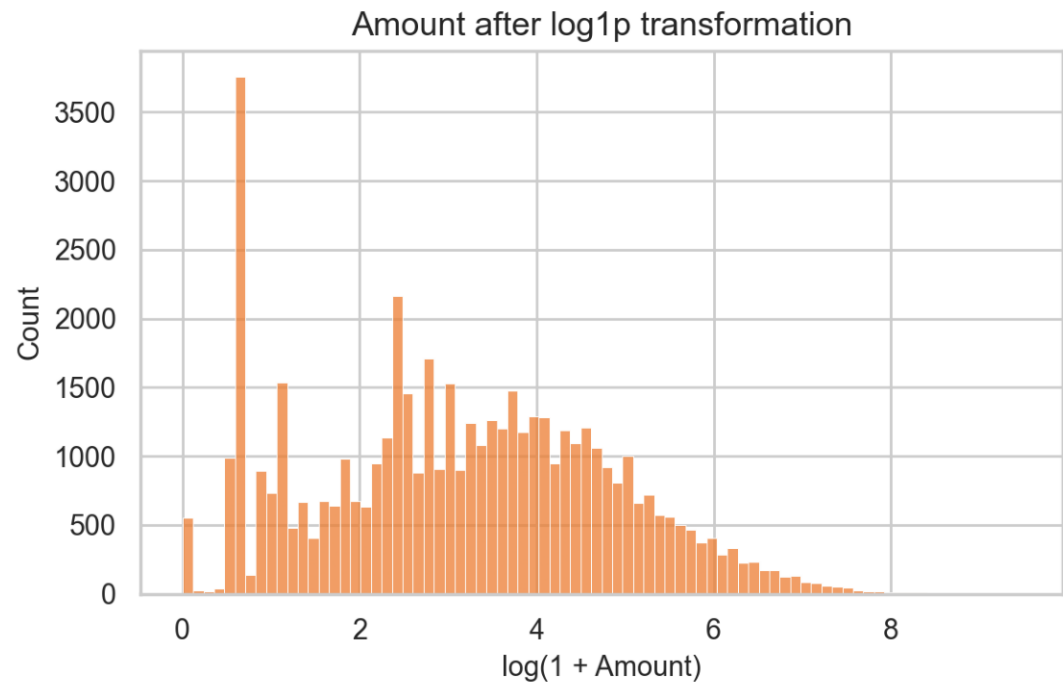
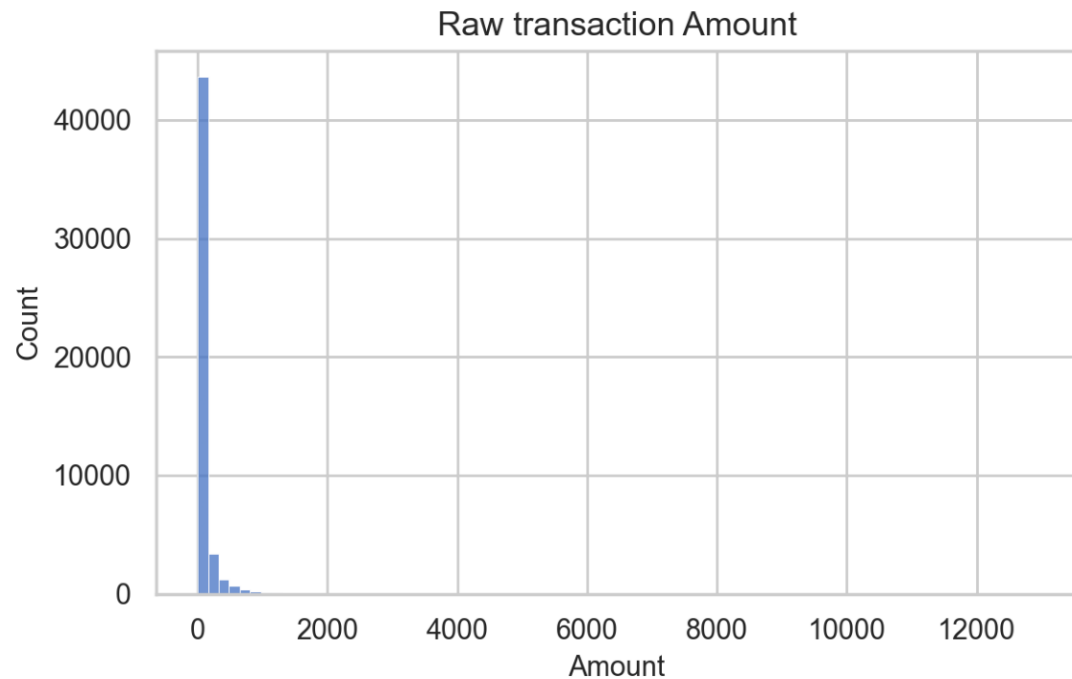
H above approximately 0.70-0.75 supports strong cluster tendency.

VAT must be interpreted jointly with Hopkins and low-dimensional embeddings. Diagonal block structure supports cluster tendency.

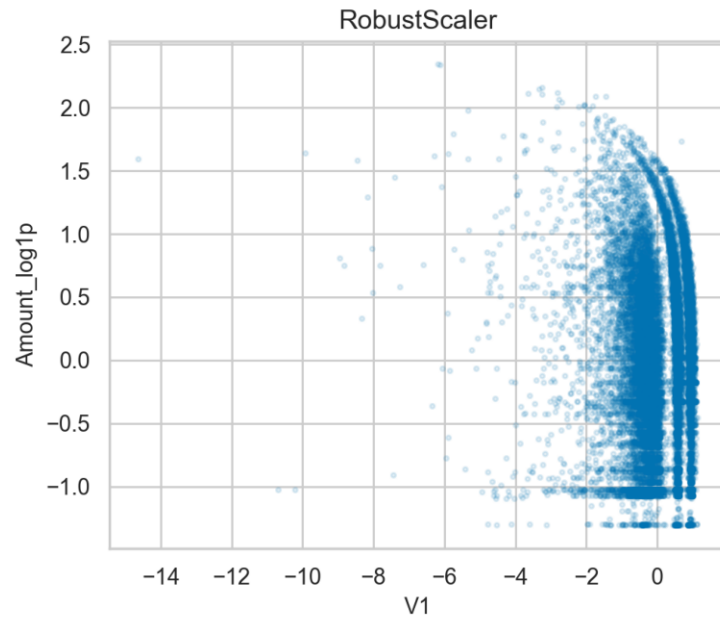
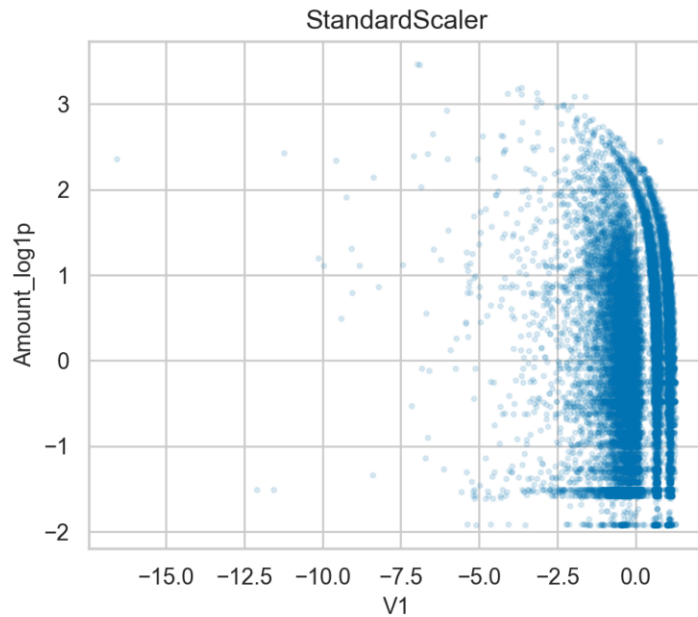
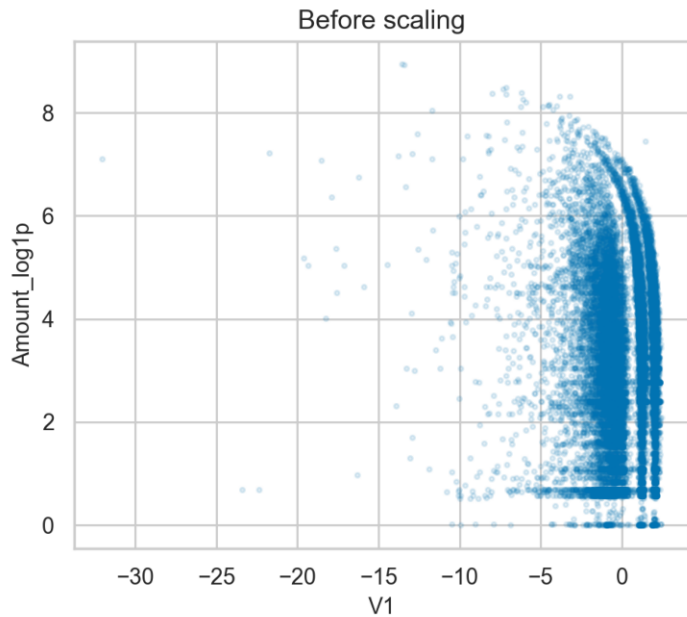
Selected Feature Statistics

	mean	std	50%	iqr	skewness	kurtosis
V1	0.0059	1.9480	0.0204	2.2320	-3.2733	32.7273
V2	-0.0041	1.6467	0.0639	1.4006	-4.6952	96.8982
V3	0.0016	1.5087	0.1800	1.9166	-2.1520	25.1865
V4	-0.0030	1.4142	-0.0222	1.5898	0.6715	2.6188
V10	-0.0014	1.0764	-0.0932	0.9892	1.2530	29.8439
V14	0.0003	0.9522	0.0502	0.9181	-1.9188	23.0415
V17	0.0002	0.8425	-0.0659	0.8829	-3.6905	93.3231
Amount_log1p	3.1538	1.6571	3.1355	2.4762	0.1614	-0.6413

Amount Transformation

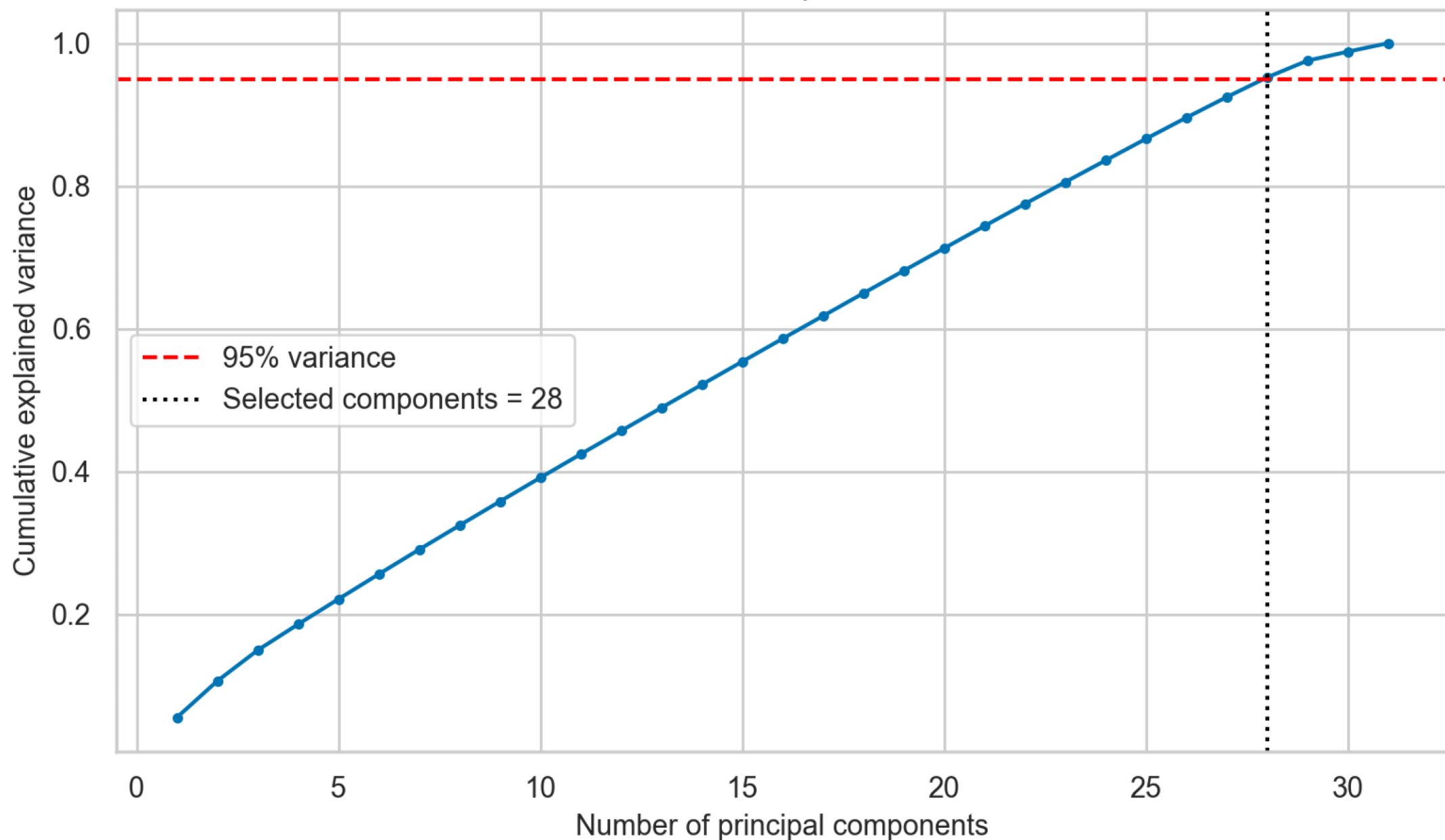


Before and After Scaling



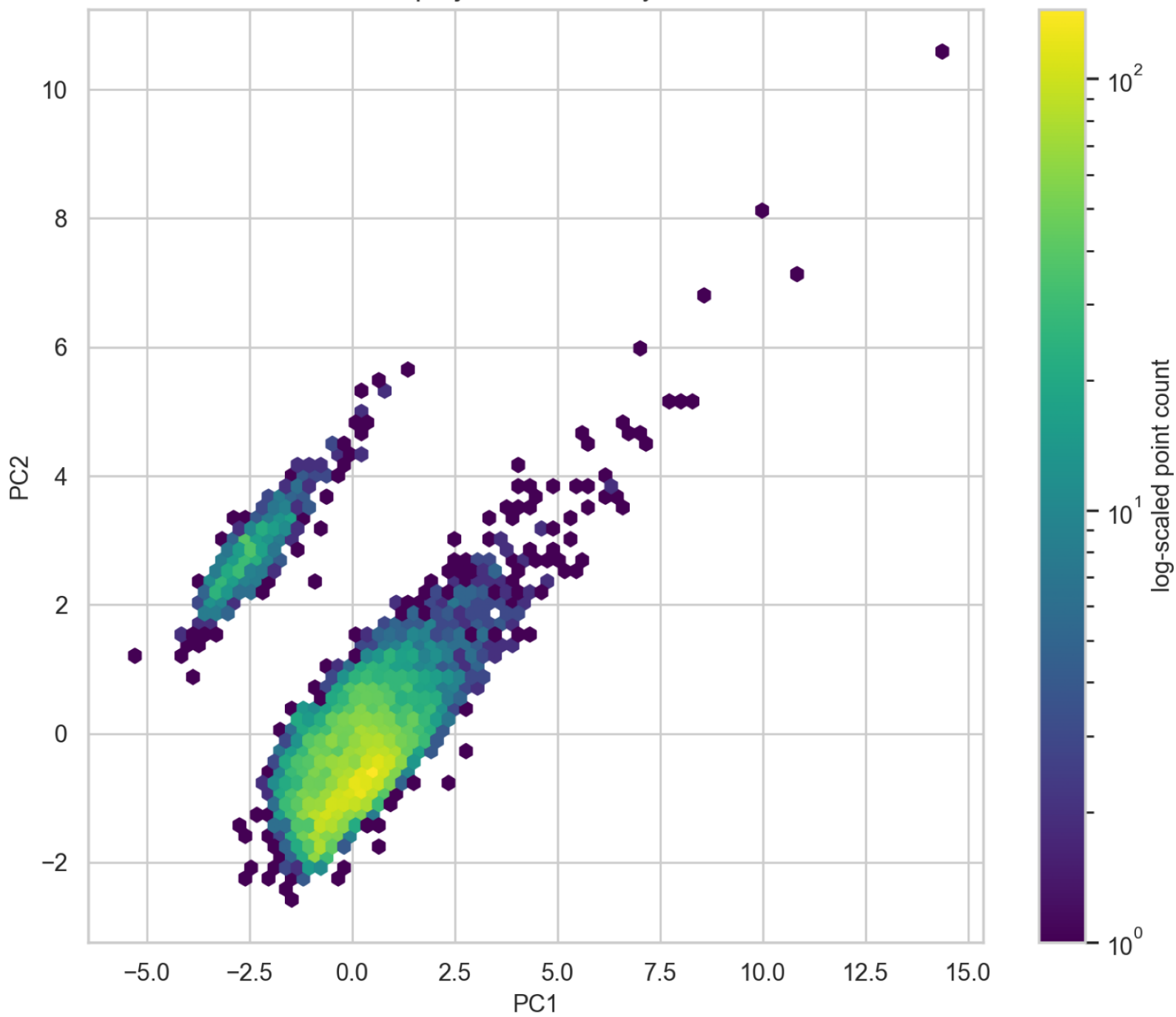
PCA Explained Variance

PCA cumulative explained variance



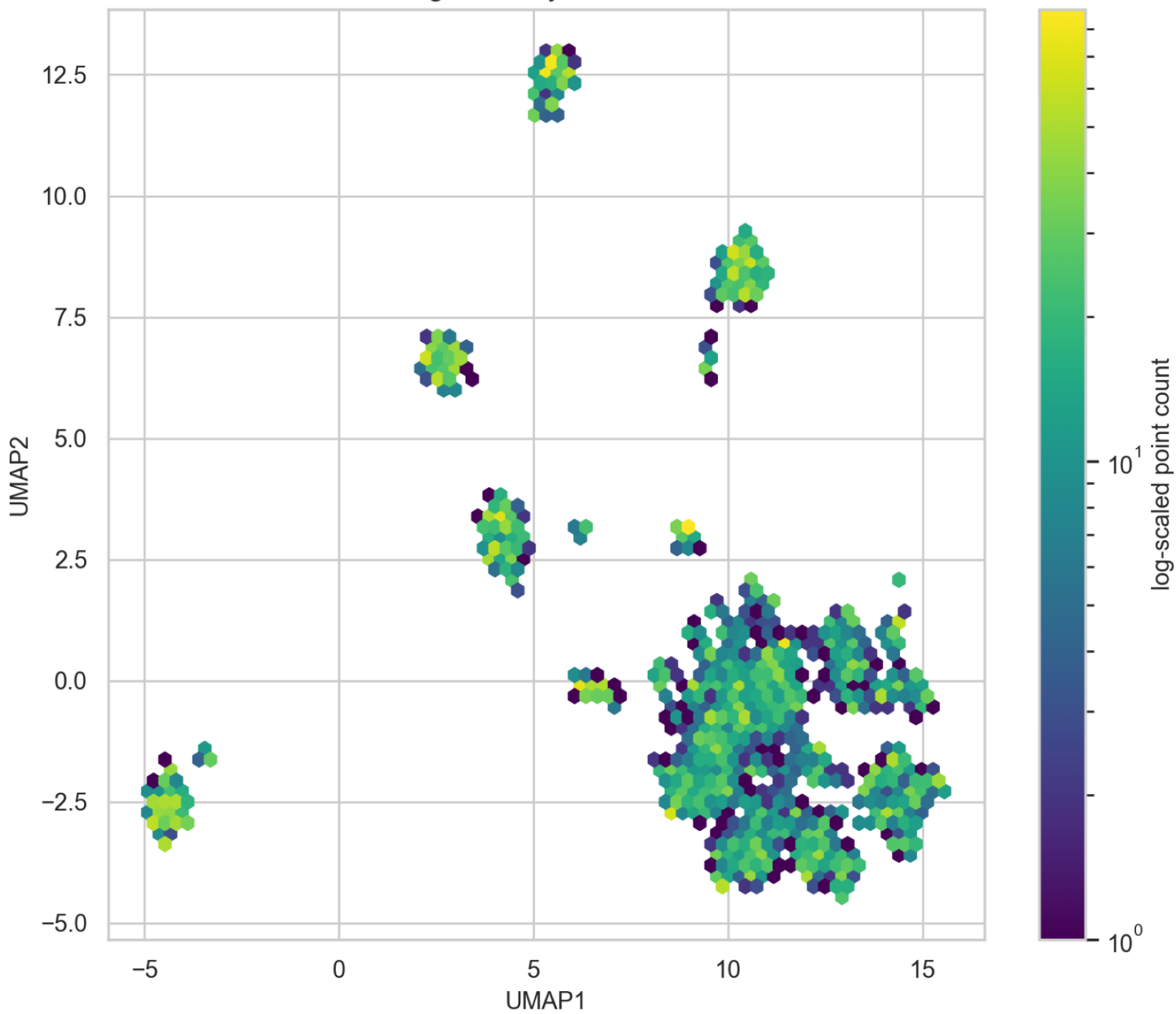
PCA Density Projection

PCA projection—density view

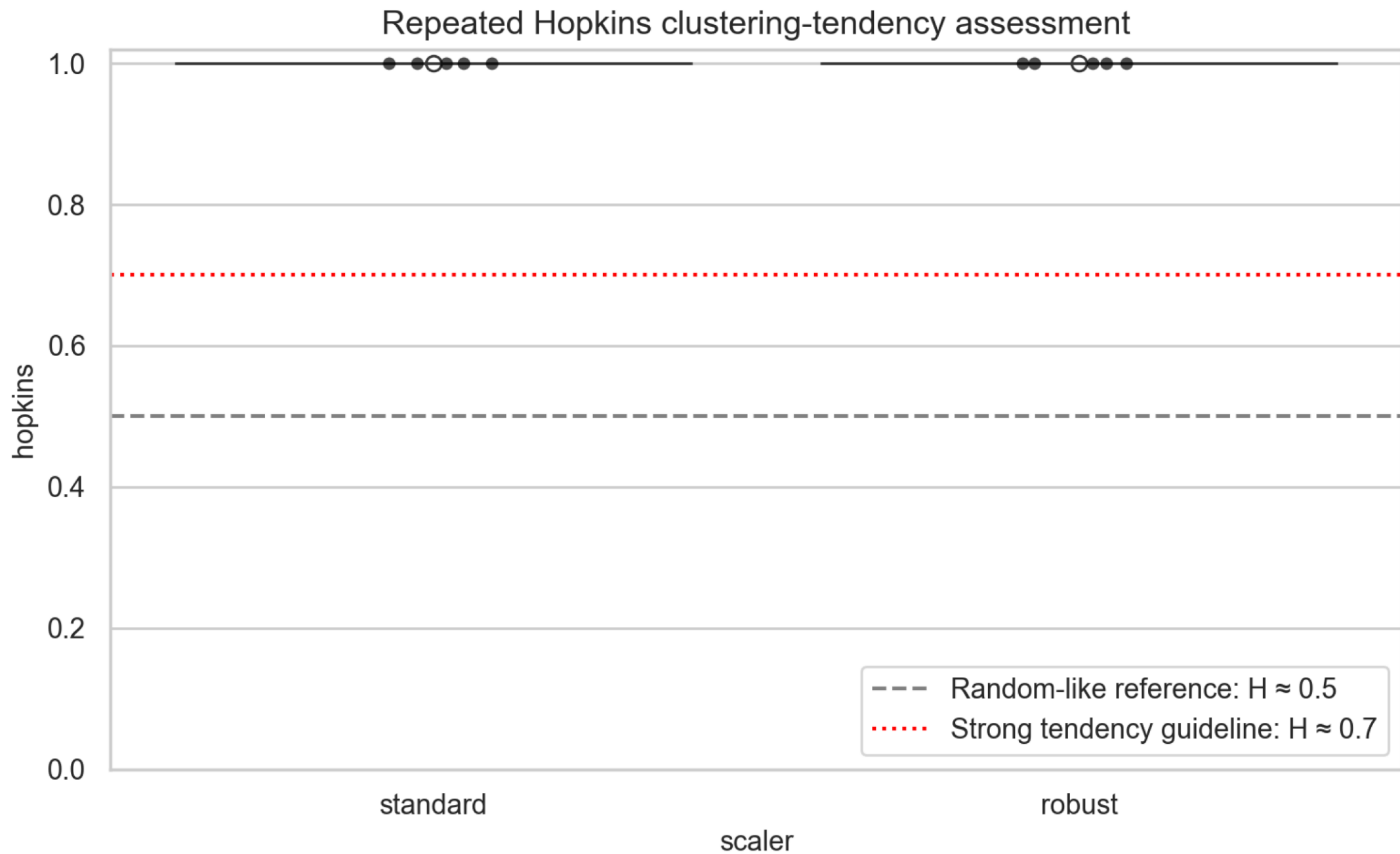


UMAP Density Projection

UMAP embedding—density view without class labels

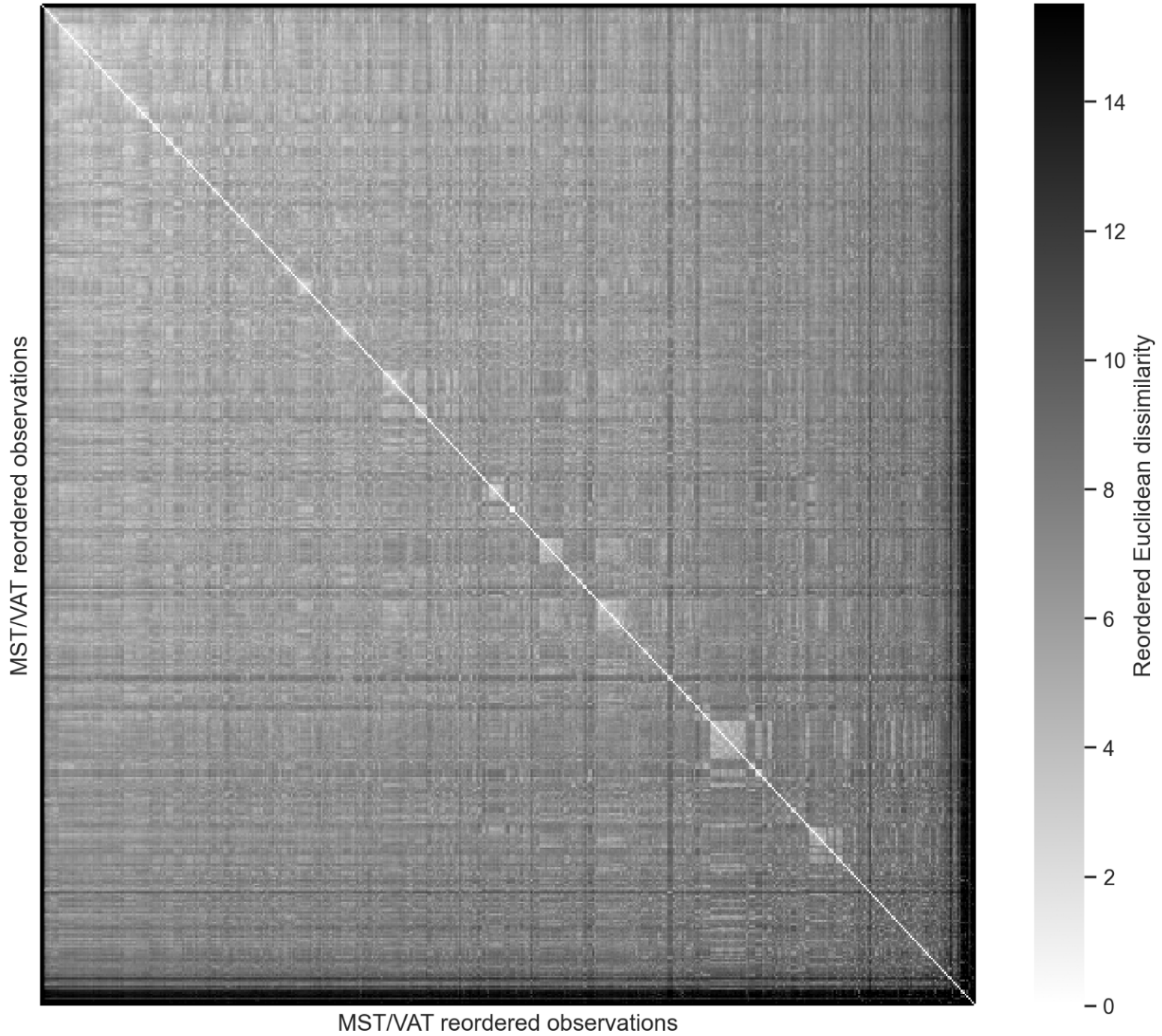


Repeated Hopkins Results



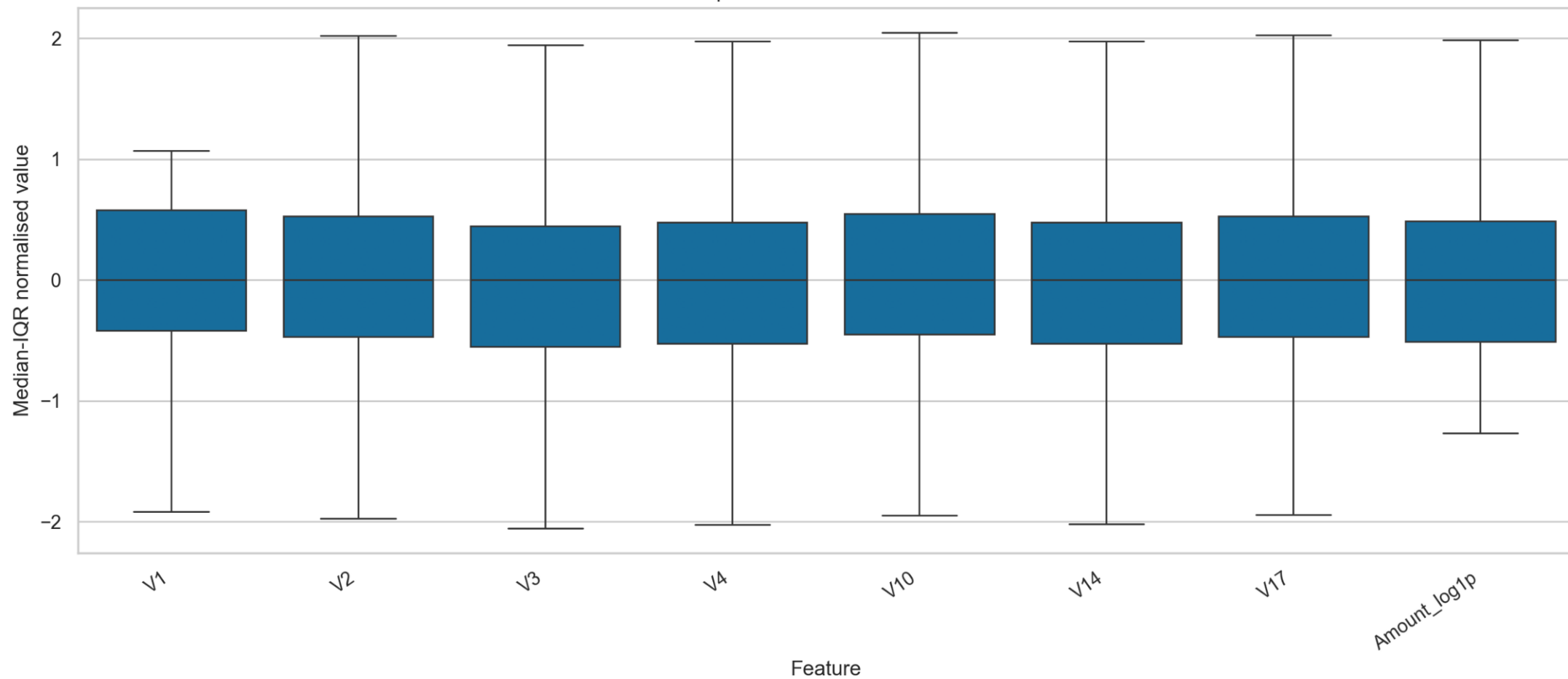
MST-Based VAT

MST-based VAT dissimilarity matrix



Selected Feature Box Plots

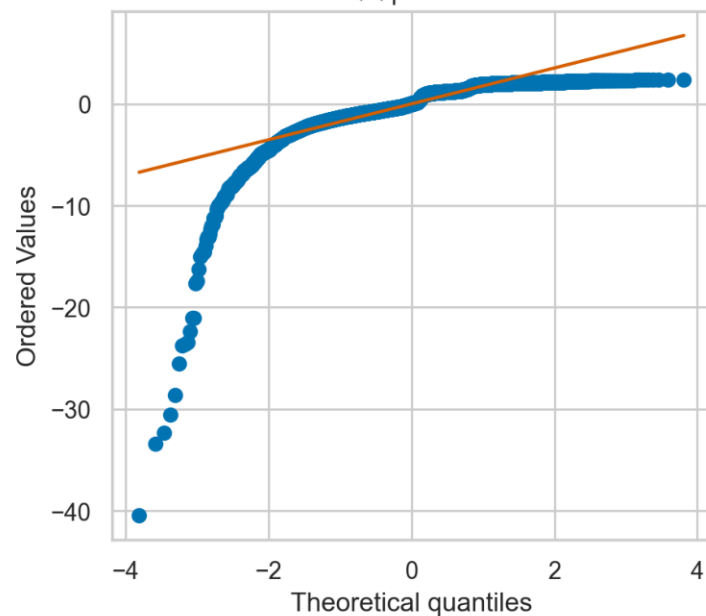
Feature box plots after median-IQR normalisation



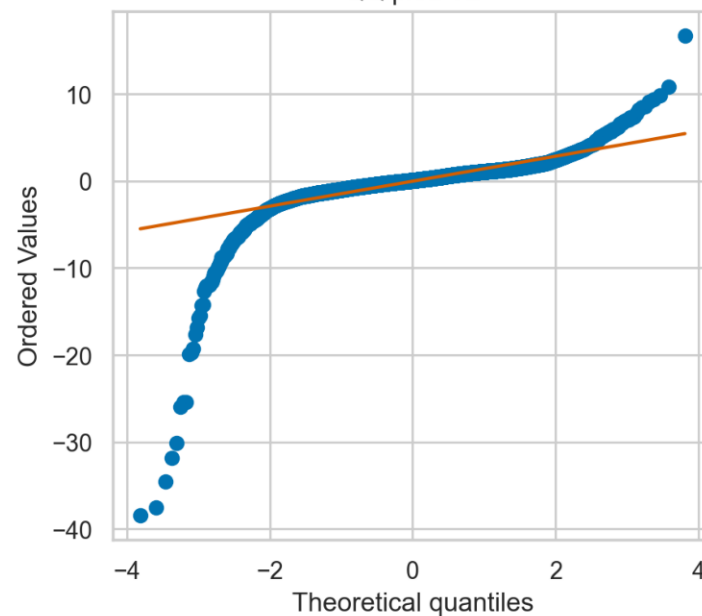
Selected Feature QQ Plots

Normal QQ plots for selected features

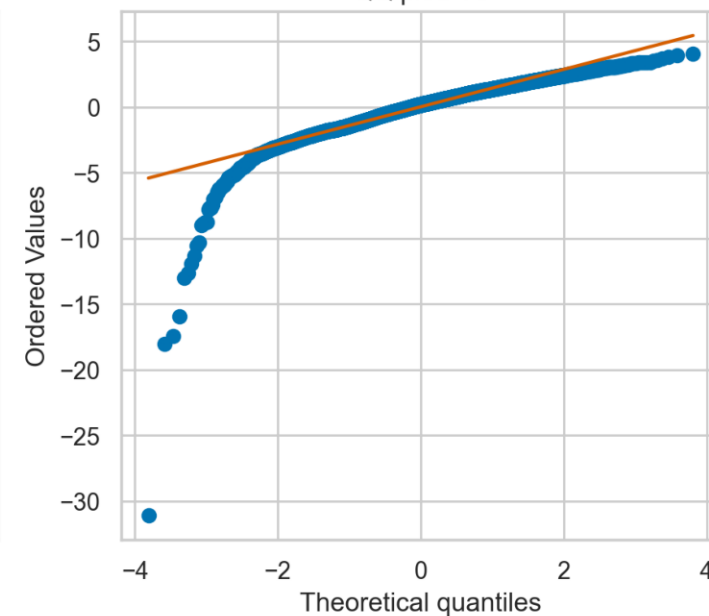
QQ plot: V1



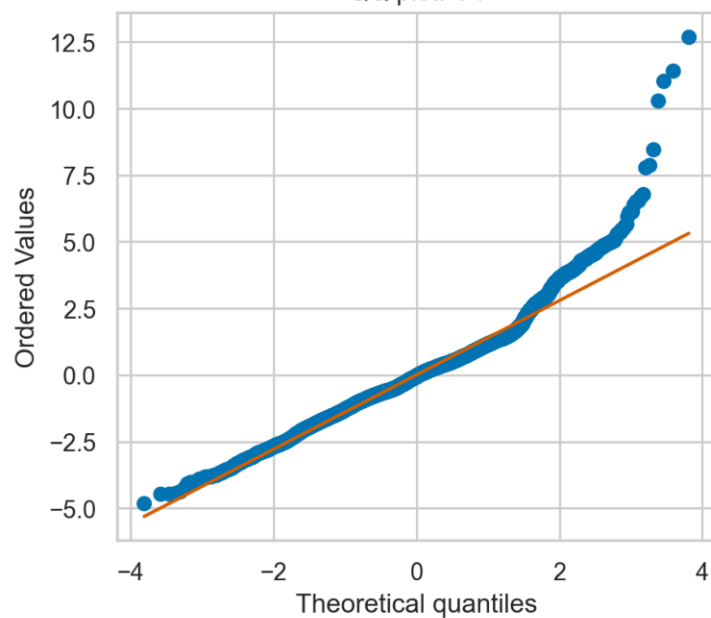
QQ plot: V2



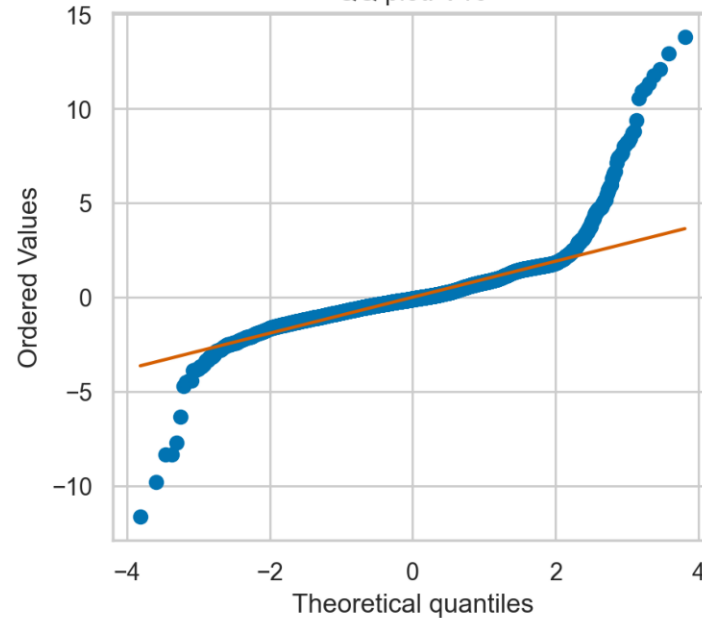
QQ plot: V3



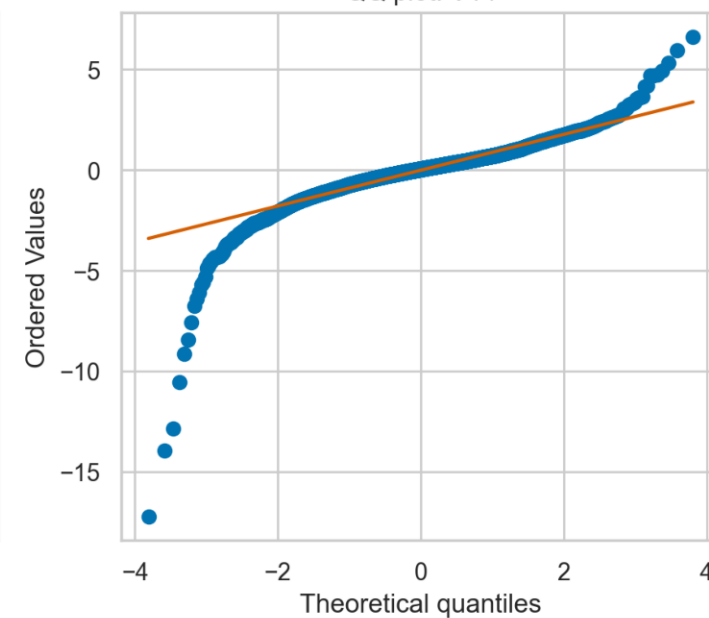
QQ plot: V4



QQ plot: V10

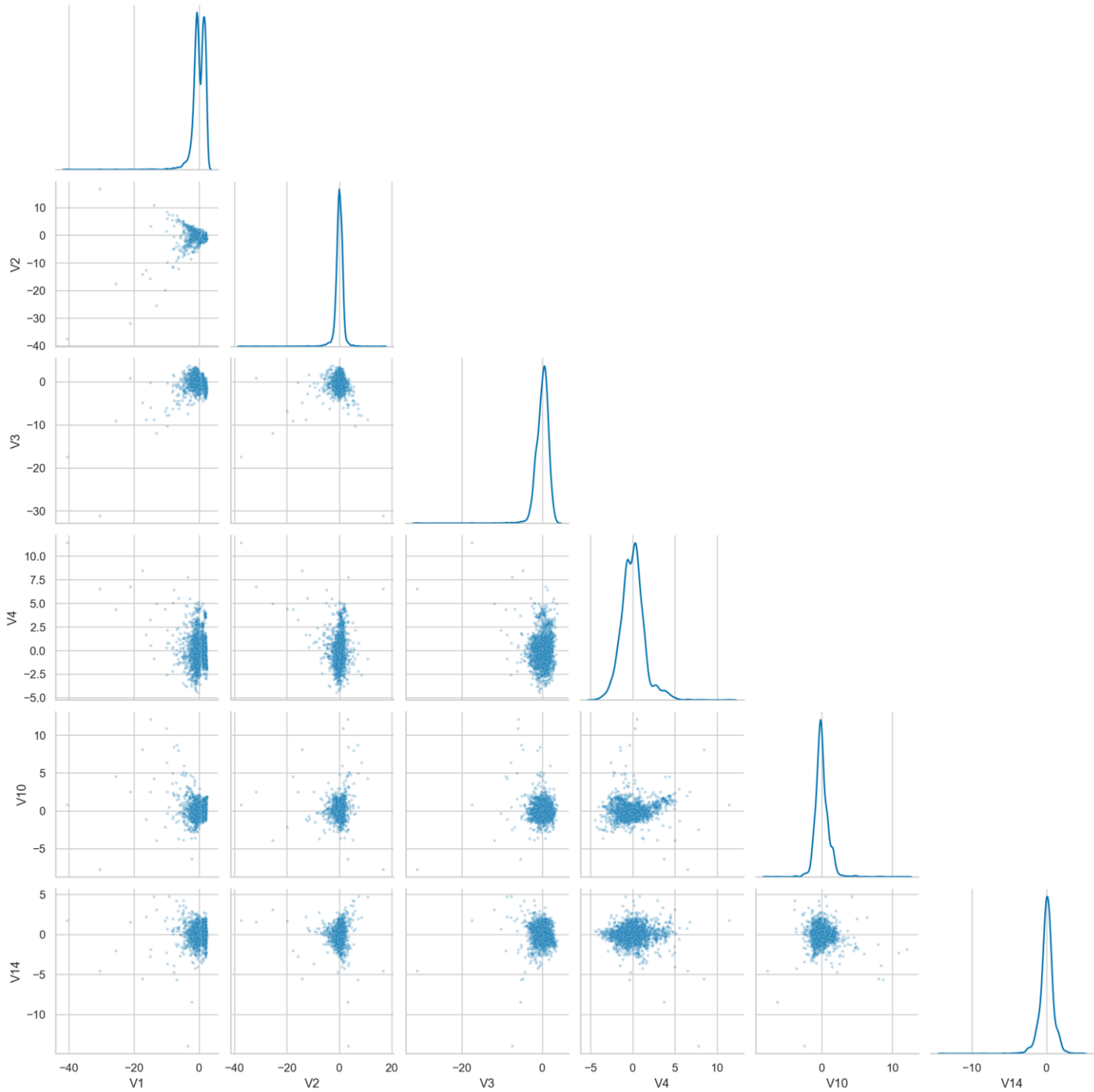


QQ plot: V14



Selected Feature Pair Plot

Scatter-plot matrix of selected features



Significance-Masked Correlation

